

Algorithmic Collusion in Networked Electricity Markets

IEEE Publication Technology, *Staff, IEEE,*

Abstract—Deregulated electricity markets may face new risks when generation firms use autonomous learning-based algorithms. Existing studies of algorithmic collusion largely rely on Q-learning agents in simplified, discrete-action market settings. However, wholesale electricity markets involve continuous-action decisions and shared transmission networks, where firms interact not only through prices and quantities but also through congestion patterns. Prior work has shown that transmission constraints can facilitate collusion by allowing strategic suppliers to decongest constrained lines and shift congestion surplus from transmission owners to generation firms. This raises an important question: can autonomous learning agents discover and sustain similar behavior in a continuous-action, network-constrained electricity market? This study investigates tacit algorithmic collusion using independent Proximal Policy Optimization (PPO) agents embedded in a market framework with DC optimal power flow (DC-OPF). We simulate a five-node transmission network with strategic generation firms that independently learn policies that maximize cumulative profit, without communication or information sharing. Using an actor-critic architecture, we track policy convergence through normalized profit gains and evaluate whether the learned strategies produce persistent prices above competitive benchmarks. Preliminary results suggest that agents can converge to stable market outcomes consistent with tacit collusion, with strategies shaped by transmission capacity limits and network topology. Deviation experiments show that unilateral deviations do not necessarily trigger price wars; instead, agents exhibit accommodation and rapid mean reversion. We further analyze how congestion, observation history length, and market concentration affect the level and stability of learned market behavior.

Index Terms—Article submission, IEEE, IEEEtran, journal, LATEX, paper, template, typesetting.

I. INTRODUCTION

AS algorithmic trading replaces human decision-making in traditional markets such as power markets, it has become more important than ever to verify that these agents are not learning to manipulate the market prices. Since these agents interact repeatedly in a similar environment, they can eventually learn sophisticated strategies by colluding tacitly and sustain prices that are higher than the competitive price. Furthermore, certain features of electricity markets, such as the barrier to entry for new firms and the dominance of incumbent power generators, exacerbate market dynamics. A higher price means that the extra profits of firms come at the expense of consumers. Moreover, this implicit collusion among the generators can be a headache for the Department of Justice, as it is very difficult to prove.

Previous work demonstrates the collusive behavior of generators using a static economic model. However, these models

cannot simulate the learning dynamics of complex AI pricing agents and also do not explain the process by which these collusive equilibria are sustained. Recent studies using multi-agent reinforcement learning in this domain have been limited to Q-learning algorithms. Even though Q-learning agents do demonstrate and explain the process by which these agents tacitly collude, there is still a significant limitation due to the discretization of state and action spaces, which is hardly the case in electricity markets. There remains a critical gap in understanding how sophisticated agents based on complex deep reinforcement learning algorithms behave in these highly critical, fast-paced, and highly sensitive electricity markets.

II. RESEARCH METHODOLOGY

A. Perfect Competition Benchmark

In the perfect competition benchmark, we assume that all market participants act as price takers. By the First Fundamental Theorem of Welfare Economics, the competitive market equilibrium coincides with the solution of a central planner maximizing the total social welfare. Social welfare is defined as the gross consumer surplus minus the total cost of generation. Crucially, this competitive model serves as the theoretical floor benchmark for evaluating the severity and impact of the subsequent collusion simulations.

The objective is to maximize social welfare subject to the physical constraints of the direct current (DC) transmission network, including the global energy balance and the thermal limits of the transmission lines. Let $\mathcal{N}$ denote the set of nodes, $\mathcal{G}$ the set of generators, and $\mathcal{L}$ the set of transmission lines. The decision variables are the demand d_i at each node $i \in \mathcal{N}$ and the generation $g_{f,i}$ for each firm f at node i .

The centralized optimization problem is formulated as follows:

$$\begin{aligned} \max_{d,g} \quad & \sum_{i \in \mathcal{N}} \left(P_{0i} d_i - \frac{1}{2} \frac{P_{0i}}{Q_{0i}} d_i^2 \right) \\ & - \sum_{(f,i) \in \mathcal{G}} \left(MC_{f,i} g_{f,i} + \frac{1}{2} QC_{f,i} g_{f,i}^2 \right) \end{aligned} \quad (1)$$

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subject to:

$$\sum_{i \in \mathcal{N}} \left(\sum_f g_{f,i} - d_i \right) = 0 \quad : \lambda \quad (2)$$

$$\sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \leq T_k \quad : \mu_k^+, \forall k \in \mathcal{L} \quad (3)$$

$$- \sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \leq T_k \quad : \mu_k^-, \forall k \in \mathcal{L} \quad (4)$$

$$g_{f,i} \leq CAP_{f,i} \quad : \rho_{f,i}, \quad \forall (f,i) \in \mathcal{G} \quad (5)$$

$$d_i \geq 0, \quad g_{f,i} \geq 0 \quad (6)$$

Where:

- $P0_i$ and $Q0_i$ are the price and quantity intercepts of the inverse demand function at node i .
- $MC_{f,i}$ and $QC_{f,i}$ are the linear and quadratic marginal cost coefficients for generator f at node i .
- $PTDF_{k,i}$ is the Power Transfer Distribution Factor mapping the net injection at node i to the active power flow on line k .
- T_k is the thermal capacity limit of transmission line k .
- $\lambda, \mu_k^+, \mu_k^-, \rho_{f,i}$ are the dual variables (Lagrange multipliers) associated with each respective constraint.

1) *Karush-Kuhn-Tucker (KKT) Conditions*: Because the objective function is a strictly concave quadratic function and the equality and inequality constraints are linear, the feasible region is convex. Therefore, the KKT conditions are both necessary and sufficient for global optimality. The Lagrangian is constructed by appending the constraints to the objective function, and the conditions are derived by taking the partial derivatives with respect to the decision variables.

The first-order stationarity conditions with respect to demand d_i and generation $g_{f,i}$, represented in complementarity form, are:

$$0 \leq d_i \perp -P0_i + \frac{P0_i}{Q0_i} d_i + \lambda - \sum_{k \in \mathcal{L}} PTDF_{k,i} (\mu_k^+ - \mu_k^-) \geq 0 \quad (7)$$

$$0 \leq g_{f,i} \perp MC_{f,i} + QC_{f,i} g_{f,i} - \lambda + \sum_{k \in \mathcal{L}} PTDF_{k,i} (\mu_k^+ - \mu_k^-) + \rho_{f,i} \geq 0 \quad (8)$$

The primal and dual feasibility, along with the complementary slackness conditions for the network and capacity

constraints, are:

$$\sum_{i \in \mathcal{N}} \left(\sum_f g_{f,i} - d_i \right) = 0 \quad (9)$$

$$0 \leq \mu_k^+ \perp T_k - \sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \geq 0, \quad \forall k \in \mathcal{L} \quad (10)$$

$$0 \leq \mu_k^- \perp T_k + \sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \geq 0, \quad \forall k \in \mathcal{L} \quad (11)$$

$$0 \leq \rho_{f,i} \perp CAP_{f,i} - g_{f,i} \geq 0, \quad \forall (f,i) \in \mathcal{G} \quad (12)$$

From the stationarity condition regarding d_i , we can extract the Locational Marginal Price (LMP) at node i , which represents the marginal benefit of electricity. If demand is strictly positive ($d_i > 0$), the LMP is defined exactly by the system's global energy balance dual minus the congestion components: $LMP_i = \lambda - \sum_{k \in \mathcal{L}} PTDF_{k,i} (\mu_k^+ - \mu_k^-)$. In this competitive setting, the stationarity condition for generation mathematically guarantees that each firm produces up to the exact point where their marginal cost of generation equates to the LMP at their respective node.

B. Joint Monopoly Benchmark

In contrast to the perfect competition baseline, the joint monopoly benchmark represents the theoretical ceiling for collusive behavior. Here, we assume all generating firms act as a single, perfectly coordinated cartel. Rather than maximizing social welfare, the central objective of the joint monopoly is to maximize the aggregate industry profit across the entire network. This requires perfectly internalizing the effect of generated quantities on the clearing prices at each node, exploiting the network's congestion patterns to restrict supply and elevate prices.

The physical constraints of the grid remain identical to the competitive case. However, the objective function shifts from gross consumer surplus to total market revenue. Because revenue is defined as $P_i d_i$, the integration factor of $\frac{1}{2}$ present in the welfare objective is removed.

The centralized monopoly optimization problem is formulated as follows:

$$\max_{d,g} \sum_{i \in \mathcal{N}} \left(P0_i d_i - \frac{P0_i}{Q0_i} d_i^2 \right) - \sum_{(f,i) \in \mathcal{G}} \left(MC_{f,i} g_{f,i} + \frac{1}{2} QC_{f,i} g_{f,i}^2 \right) \quad (13)$$

subject to:

$$\sum_{i \in \mathcal{N}} \left(\sum_f g_{f,i} - d_i \right) = 0 \quad : \lambda \quad (14)$$

$$\sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \leq T_k \quad : \mu_k^+, \forall k \in \mathcal{L} \quad (15)$$

$$- \sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \leq T_k \quad : \mu_k^-, \forall k \in \mathcal{L} \quad (16)$$

$$g_{f,i} \leq CAP_{f,i} \quad : \rho_{f,i}, \quad \forall (f,i) \in \mathcal{G} \quad (17)$$

$$d_i \geq 0, \quad g_{f,i} \geq 0 \quad (18)$$

1) *Karush-Kuhn-Tucker (KKT) Conditions:* The joint monopoly formulation remains a strictly concave quadratic program subject to linear constraints, preserving convexity. Thus, the KKT conditions remain necessary and sufficient for global optimality.

The critical departure from perfect competition is found in the first-order stationarity condition for demand. Taking the partial derivative of the revenue function with respect to d_i yields the marginal revenue rather than the price, introducing a factor of 2 into the slope term.

The first-order stationarity conditions, represented in complementarity form, are:

$$0 \leq d_i \perp -P0_i + 2 \frac{P0_i}{Q0_i} d_i + \lambda - \sum_{k \in \mathcal{L}} PTDF_{k,i} (\mu_k^+ - \mu_k^-) \geq 0 \quad (19)$$

$$0 \leq g_{f,i} \perp MC_{f,i} + QC_{f,i} g_{f,i} - \lambda + \sum_{k \in \mathcal{L}} PTDF_{k,i} (\mu_k^+ - \mu_k^-) + \rho_{f,i} \geq 0 \quad (20)$$

The primal and dual feasibility, along with the complementary slackness conditions, govern the identical physical network and remain unchanged from the competitive benchmark:

$$\sum_{i \in \mathcal{N}} \left(\sum_f g_{f,i} - d_i \right) = 0 \quad (21)$$

$$0 \leq \mu_k^+ \perp T_k - \sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \geq 0, \quad \forall k \in \mathcal{L} \quad (22)$$

$$0 \leq \mu_k^- \perp T_k + \sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \geq 0, \quad \forall k \in \mathcal{L} \quad (23)$$

$$0 \leq \rho_{f,i} \perp CAP_{f,i} - g_{f,i} \geq 0, \quad \forall (f,i) \in \mathcal{G} \quad (24)$$

C. Nash-Cournot Equilibrium Benchmark

The Nash-Cournot equilibrium establishes the individually rational, non-cooperative baseline for the market. Unlike the perfect competition benchmark where firms are price takers,

under the Cournot assumption, generation firms recognize that their individual production quantities directly affect the local nodal prices (Here we assume an exogenous ISO model)

Because the firms and the ISO pursue different, independent objectives (profit maximization versus welfare maximization), the equilibrium is found by formulating a Linear Complementarity Problem (LCP). This is achieved by deriving the first-order Karush-Kuhn-Tucker (KKT) conditions for each individual firm's profit maximization problem and stacking them alongside the KKT conditions of the ISO's market-clearing problem.

For each firm f , the marginal revenue explicitly accounts for the downward slope of the inverse demand curve, $\frac{P0_i}{Q0_i}$. The stationarity and capacity conditions for generation become:

$$0 \leq g_{f,i} \perp - \left(P0_i - \frac{P0_i}{Q0_i} d_i \right) + \frac{P0_i}{Q0_i} g_{f,i} + MC_{f,i} + QC_{f,i} g_{f,i} + \rho_{f,i} \geq 0 \quad (25)$$

$$0 \leq \rho_{f,i} \perp CAP_{f,i} - g_{f,i} \geq 0, \quad \forall (f,i) \in \mathcal{G} \quad (26)$$

The ISO independently dispatches the grid to maximize welfare, treating generation as given. Using the same dual variables as the competitive model, the ISO's conditions are:

$$\left(P0_i - \frac{P0_i}{Q0_i} d_i \right) - \lambda + \sum_{k \in \mathcal{L}} PTDF_{k,i} (\mu_k^+ - \mu_k^-) = 0, \quad \forall i \in \mathcal{N} \quad (27)$$

$$\sum_{i \in \mathcal{N}} \left(\sum_f g_{f,i} - d_i \right) = 0 \quad (28)$$

$$0 \leq \mu_k^+ \perp T_k - \sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \geq 0, \quad \forall k \in \mathcal{L} \quad (29)$$

$$0 \leq \mu_k^- \perp T_k + \sum_{i \in \mathcal{N}} PTDF_{k,i} \left(\sum_f g_{f,i} - d_i \right) \geq 0, \quad \forall k \in \mathcal{L} \quad (30)$$

The critical distinction in this LCP formulation is the addition of the $\frac{P0_i}{Q0_i} g_{f,i}$ term in the generation stationarity condition. This outcome establishes the true "no-collusion" floor for our Δ index.

D. Numerical Benchmark Comparison

To establish a quantitative scale for the subsequent experiments, the perfect competition, Nash-Cournot, and joint monopoly programs are solved on the five-node test network using a convex programming and LCP solver. The resulting equilibria provide the competitive floor, the individually rational non-cooperative baseline, and the collusive ceiling against which the converged PPO policies are evaluated. Tables I–V summarize the comparison along four dimensions: aggregate market outcomes, per-firm profits, plant-level generation, and nodal locational marginal prices.

The numerical results highlight the severe impact of structural market power even in the absence of explicit collusion.

TABLE I
HEADLINE MARKET METRICS UNDER THE THREE BENCHMARKS

Metric	Competitive	Cournot-Nash	Monopoly	% Δ (C $\rightarrow$ N)	% Δ (N $\rightarrow$ M)	% Δ (C $\rightarrow$ M)
Avg. LMP (\$/MWh)	23.49	29.35	28.86	+24.9%	-1.7%	+22.9%
Total generation (MW)	282.43	176.24	198.68	-37.6%	+12.7%	-29.6%
Total industry profit (\$/step)	1085.19	2028.32	2267.83	+87.0%	+11.8%	+109.0%

TABLE II
PER-FIRM PROFIT AT EQUILIBRIUM

Firm	Comp. (\$)	Nash (\$)	Mono. (\$)	vs. Comp.	vs. Nash
Firm 0	1051.22	1552.61	1949.08	1.85 $\times$	1.26 $\times$
Firm 1	33.98	475.71	318.75	9.38 $\times$	0.67 $\times$
Total	1085.19	2028.32	2267.83	2.09 $\times$	1.12 $\times$

TABLE III
PLANT UTILIZATION DROP VS. CAPACITY

Plant	Comp. Drop	Nash Drop	Mono. Drop
0	0.0%	-50.1%	-25.9%
1	0.0%	0.0%	0.0%
2	-17.6%	-48.6%	-62.5%

TABLE IV
GENERATION PER PLANT (THREE BENCHMARKS)

Plant	Firm	Node	Cap (MW)	Comp. (MW)	Nash (MW)	Monopoly (MW)	Δ vs. Comp. (MW)
0	0	0	150	150.00	74.84	111.18	-75.16
1	0	1	50	50.00	50.00	50.00	0.00
2	1	1	100	82.43	51.41	37.50	-31.02
Total	-	-	300	282.43	176.24	198.68	-106.19

TABLE V
NODAL LOCATIONAL MARGINAL PRICES (THREE BENCHMARKS)

Node	Role	Comp. (\$/MWh)	Nash (\$/MWh)	Mono. (\$/MWh)	Δ C $\rightarrow$ N	Δ N $\rightarrow$ M
0	Supply (Firm 0)	22.40	28.47	28.61	+6.07	+0.14
1	Supply (Firms 0, 1)	18.82	27.51	26.69	+8.69	-0.82
2	Load only	28.00	29.43	28.00	+1.43	-1.43
3	Load only	29.00	29.43	29.00	+0.43	-0.43
4	Load only	34.00	34.00	34.00	0.00	0.00

Moving from perfect competition to the Nash-Cournot baseline, total industry profit nearly doubles as firms unilaterally withhold capacity, dropping total generation from 282.43 MW to 176.24 MW. This non-cooperative withholding artificially congests the network and causes Locational Marginal Prices (LMPs) to spike at the supply-side nodes.

Crucially, the transition from the non-cooperative Nash equilibrium to the joint monopoly reveals the highly asymmetric nature of full network coordination. To maximize aggregate industry profit, the monopolist actually increases total generation slightly relative to the Nash baseline (from 176.24 MW to 198.68 MW). However, it achieves this by drastically shifting production away from Firm 1's plant at Node 1. Consequently, Firm 1's profit falls from \$475.71 under the Nash equilibrium to \$318.75 under the joint monopoly.

E. Collusion Metric

To quantify tacit collusion, we define a normalized profit gain index, Δ . A naive approach calculates this per-firm using perfect competition as the baseline floor:

$$\Delta_f^{\text{flawed}} = \frac{\bar{\pi}_f - \pi_f^{\text{Comp}}}{\pi_f^{\text{Mono}} - \pi_f^{\text{Comp}}} \quad (31)$$

This fails in transmission-constrained networks because both the firms will play at least their Nash Cournot baseline. Moreover, joint monopolies maximize aggregate profit by forcing specific firms to restrict output below their individually

rational baseline. This creates a negative denominator for those firms, rendering the index mathematically unstable for independent agents lacking side-payments.

Therefore, we establish the non-cooperative Nash-Cournot equilibrium as the true floor and compute the metric on a combined basis:

$$\Delta_{\text{comb}} = \frac{\sum_f (\bar{\pi}_f - \pi_f^{\text{Nash}})}{\sum_f (\pi_f^{\text{Mono}} - \pi_f^{\text{Nash}})} \quad (32)$$

Crucially, Δ_{comb} also dictates our PPO early-stopping criteria. Training terminates when the absolute change in Δ_{Ind} between consecutive PPO updates remains below a strict tolerance threshold for a sustained patience window, guaranteeing the agents have naturally converged to a stable equilibrium.

F. Multi-Agent PPO Algorithm

To simulate the learning dynamics of strategic generation firms, we employ a Multi-Agent Independent Proximal Policy Optimization (IPPO) framework. Algorithm 1 outlines the complete training loop. The process is divided into four distinct phases: (1) experience collection via interaction with the ISO's DC-OPF market clearing, (2) advantage estimation using Generalized Advantage Estimation (GAE), (3) clipped network updates with an entropy bonus to encourage exploration, and (4) an external convergence supervisor. Instead of using a fixed number of timesteps, the supervisor continuously monitors the combined collusion index (Δ_{comb}) and halts

the simulation once the aggregate market profit successfully stabilizes.

Algorithm 1 Tacit Collusion with PPO

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1: Input: Hyperparameters  $\gamma, \lambda, \epsilon, \alpha, c_1, c_2, E, M, T, \tau, N$ .
2: Initialize: Actor  $\theta_{f,0}$  and Critic  $\phi_{f,0}$  for firms  $f \in \{0, 1\}$ .
3: Initialize: Welford stats  $\mu_f = \mathbf{0}, \sigma_f^2 = \mathbf{1}$  for  $f \in \{0, 1\}$ .
4: Given: Cournot-Nash profit  $\pi_{N,f}$ , Monopoly profit  $\pi_{M,f}$ .
5: for update  $k = 0, 1, 2, \dots$  do
6:   // Phase 1: Rollout & ISO Market Clearing
7:   for  $t = 0, \dots, T - 1$  do
8:     for each firm  $f \in \{0, 1\}$  do
9:        $\mu_f, \sigma_f^2 \leftarrow \text{UpdateWelford}(\mu_f, \sigma_f^2, s_{f,t})$ 
10:       $\tilde{s}_{f,t} = (s_{f,t} - \mu_f) / \sqrt{\sigma_f^2 + 10^{-8}}$ 
11:      Sample action:  $a_{f,t} \sim \pi_{\theta_{f,k}}(\cdot | \tilde{s}_{f,t})$ 
12:       $l_{f,t}^{\text{old}} = \log \pi_{\theta_{f,k}}(a_{f,t} | \tilde{s}_{f,t})$ 
13:    end for
14:    ISO clears DC-OPF to compute nodal  $LMP_{i,t}$ .
15:    Calculate exact profit  $r_{f,t}$  for each firm.
16:  end for
17:  // Phase 2: Generalized Advantage Estimation (GAE)
18:  for each firm  $f \in \{0, 1\}$  do
19:     $\delta_{f,t} = r_{f,t} + \gamma V_{\phi_{f,k}}(\tilde{s}_{f,t+1}) - V_{\phi_{f,k}}(\tilde{s}_{f,t})$ 
20:     $\hat{A}_{f,t} = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l \delta_{f,t+l}$ 
21:     $G_{f,t} = \hat{A}_{f,t} + V_{\phi_{f,k}}(\tilde{s}_{f,t})$ 
22:    Normalize  $\hat{A}_{f,t}$  across the rollout buffer.
23:  end for
24:  // Phase 3: Clipped Network Updates
25:  for each firm  $f \in \{0, 1\}$  do
26:    for epoch  $= 1, \dots, E$  do
27:      for minibatch  $\mathcal{B} \subset \{0, \dots, T - 1\}$  of size  $M$  do
28:         $\rho_{f,t}(\theta) = \exp(\log \pi_{\theta}(a_{f,t} | \tilde{s}_{f,t}) - l_{f,t}^{\text{old}})$ 
29:         $L_f^{\text{CLIP}} = \frac{1}{M} \sum_{t \in \mathcal{B}} \min(\rho_{f,t} \hat{A}_{f,t},$ 
30:           $\text{clip}(\rho_{f,t}, 1 - \epsilon, 1 + \epsilon) \hat{A}_{f,t})$ 
31:         $L_f^{\text{VF}} = \frac{1}{M} \sum_{t \in \mathcal{B}} (V_{\phi}(\tilde{s}_{f,t}) - G_{f,t})^2$ 
32:         $S_f = \frac{1}{M} \sum_{t \in \mathcal{B}} \mathcal{H}(\pi_{\theta}(\cdot | \tilde{s}_{f,t}))$ 
33:         $L_f^{\text{TOTAL}} = -L_f^{\text{CLIP}} + c_1 L_f^{\text{VF}} - c_2 S_f$ 
34:        Update  $\theta, \phi$  via Adam using  $\nabla L_f^{\text{TOTAL}}$ .
35:      end for
36:    end for
37:    Clear rollout buffer for firm  $f$ .
38:  end for
39:  // Phase 4: Convergence Supervisor
40:  Compute step profit  $\bar{\pi}_{f,k} = \frac{1}{T} \sum_{t=0}^{T-1} r_{f,t}$  for each  $f$ .
41:   $\Delta_{\text{comb},k} = \frac{\sum_f (\bar{\pi}_{f,k} - \pi_{N,f})}{\sum_f (\pi_{M,f} - \pi_{N,f})}$ 
42:  if  $|\Delta_{\text{comb},k} - \Delta_{\text{comb},k-1}| < \tau$  for  $N$  updates then
43:    Terminate (Tacit Collusion Equilibrium Reached).
44:  end if
45: end for

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G. Simulation Results

The following figures illustrate the performance and convergence of the PPO agents during the algorithmic collusion simulations.

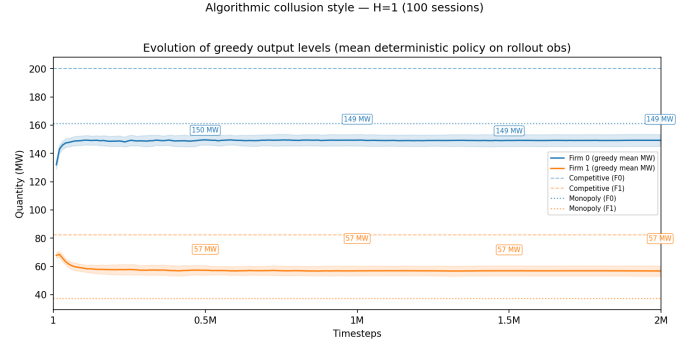


Fig. 1. Evolution of greedy output levels (mean deterministic policy on rollout observations) for $H = 1$ over 100 sessions.

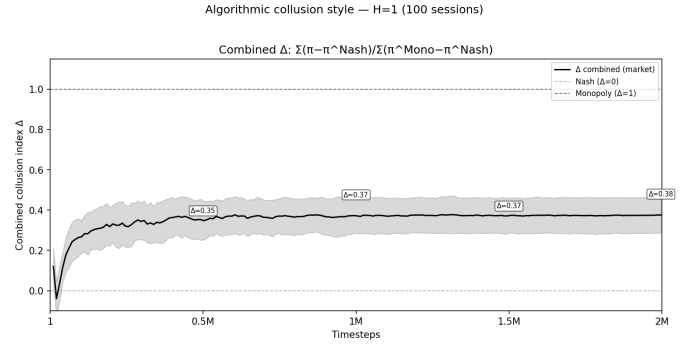


Fig. 2. Combined collusion index Δ convergence over 2 million timesteps, bounded by the Nash ($\Delta = 0$) and Monopoly ($\Delta = 1$) benchmarks.

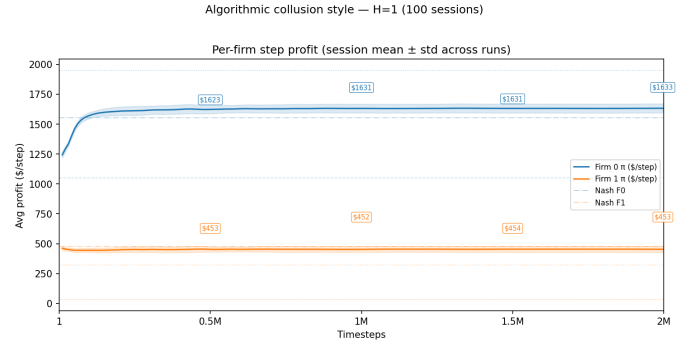


Fig. 3. Per-firm step profit (session mean $\pm$ std across runs) compared against the non-cooperative Nash baseline.

III. WHERE TO GET LATEX HELP — USER GROUPS

The following online groups are helpful to beginning and experienced LATEX users. A search through their archives can provide many answers to common questions.

<http://www.latex-community.org/>
<https://tex.stackexchange.com/>

IV. OTHER RESOURCES

See [1]–[5] for resources on formatting math into text and additional help in working with LATEX.

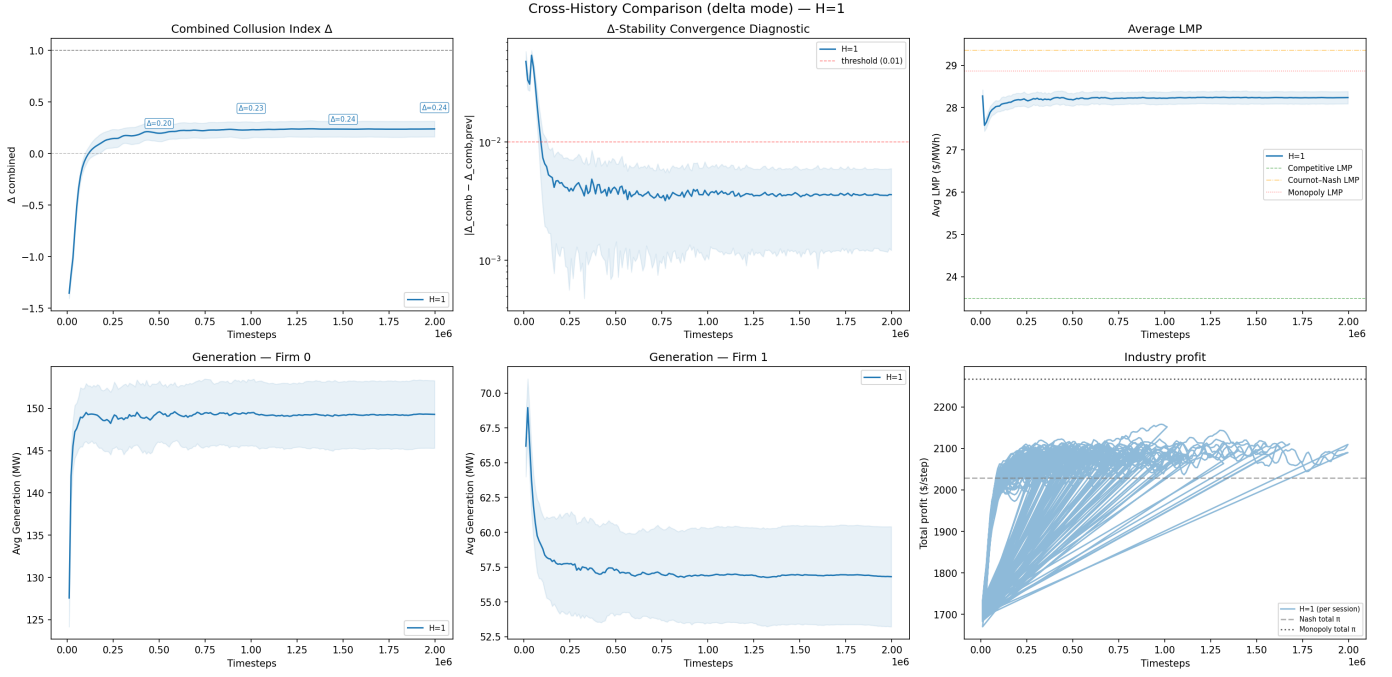


Fig. 4. Cross-history comparison for $H = 1$, detailing the combined collusion index, stability convergence diagnostic, average LMP, per-firm generation, and total industry profit.

V. TEXT

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$$x = \sum_{i=0}^n 2iQ. \quad (33)$$

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VI. SOME COMMON ELEMENTS

A. Sections and Subsections

Enumeration of section headings is desirable, but not required. When numbered, please be consistent throughout the

article, that is, all headings and all levels of section headings in the article should be enumerated. Primary headings are designated with Roman numerals, secondary with capital letters, tertiary with Arabic numbers; and quaternary with lowercase letters. Reference and Acknowledgment headings are unlike all other section headings in text. They are never enumerated. They are simply primary headings without labels, regardless of whether the other headings in the article are enumerated.

B. Citations to the Bibliography

The coding for the citations is made with the L^AT_EX `\cite` command. This will display as: see [1].

For multiple citations code as follows: `\cite{ref1,ref2,ref3}` which will produce [1]–[3]. For reference ranges that are not consecutive code as `\cite{ref1,ref2,ref3,ref9}` which will produce [1]–[3], [9]

C. Lists

In this section, we will consider three types of lists: simple unnumbered, numbered, and bulleted. There have been many options added to IEEEtran to enhance the creation of lists. If your lists are more complex than those shown below, please refer to the original “IEEEtran_HOWTO.pdf” for additional options.

A plain unnumbered list:

bare_jrnl.tex
bare_conf.tex
bare_jrnl_compsoc.tex
bare_conf_compsoc.tex
bare_jrnl_comsoc.tex

A simple numbered list:

- 1) bare_jrnl.tex
- 2) bare_conf.tex
- 3) bare_jrnl_compsoc.tex
- 4) bare_conf_compsoc.tex
- 5) bare_jrnl_comsoc.tex

A simple bulleted list:

- bare_jrnl.tex
- bare_conf.tex
- bare_jrnl_compsoc.tex
- bare_conf_compsoc.tex
- bare_jrnl_comsoc.tex

D. Figures

Fig. 1 is an example of a floating figure using the `graphicx` package. Note that `\label` must occur AFTER (or within) `\caption`. For figures, `\caption` should occur after the `\includegraphics`.

Fig. 2(a) and 2(b) is an example of a double column floating figure using two subfigures. (The `subfig.sty` package must be loaded for this to work.) The subfigure `\label` commands are set within each `subfloat` command, and the `\label` for the overall figure must come after `\caption`. `\hfil` is used



Fig. 5. Simulation results for the network.

TABLE VI
AN EXAMPLE OF A TABLE

One	Two
Three	Four

as a separator to get equal spacing. The combined width of all the parts of the figure should do not exceed the text width or a line break will occur.

Note that often IEEE papers with multi-part figures do not place the labels within the image itself (using the optional argument to `\subfloat[]`), but instead will reference/describe all of them (a), (b), etc., within the main caption. Be aware that for `subfig.sty` to generate the (a), (b), etc., subfigure labels, the optional argument to `\subfloat` must be present. If a subcaption is not desired, leave its contents blank, e.g., `\subfloat[]`.

VII. TABLES

Note that, for IEEE-style tables, the `\caption` command should come BEFORE the table. Table captions use title case. Articles (a, an, the), coordinating conjunctions (and, but, for, or, nor), and most short prepositions are lowercase unless they are the first or last word. Table text will default to `\footnotesize` as the IEEE normally uses this smaller font for tables. The `\label` must come after `\caption` as always.

VIII. ALGORITHMS

Algorithms should be numbered and include a short title. They are set off from the text with rules above and below the title and after the last line.



Fig. 6. Dae. Ad quatur autat ut porepel itemoles dolor autem fuga. Bus quia con nessunti as remo di quatus non perum que nimus. (a) Case I. (b) Case II.

Algorithm 2 Weighted Tanimoto ELM.

```

TRAIN( $\mathbf{X}\mathbf{T}$ )
  select randomly  $W \subset \mathbf{X}$ 
   $N_t \leftarrow |\{i : \mathbf{t}_i = \mathbf{t}\}|$  for  $\mathbf{t} = -1, +1$ 
   $B_i \leftarrow \sqrt{\text{MAX}(N_{-1}, N_{+1})/N_{\mathbf{t}_i}}$  for  $i = 1, \dots, N$ 
   $\hat{\mathbf{H}} \leftarrow B \cdot (\mathbf{X}^T \mathbf{W}) / (\|\mathbf{X}\| + \|\mathbf{W}\| - \mathbf{X}^T \mathbf{W})$ 
   $\beta \leftarrow (I/C + \hat{\mathbf{H}}^T \hat{\mathbf{H}})^{-1} (\hat{\mathbf{H}}^T B \cdot \mathbf{T})$ 
  return  $\mathbf{W}, \beta$ 

PREDICT( $\mathbf{X}$ )
   $\mathbf{H} \leftarrow (\mathbf{X}^T \mathbf{W}) / (\|\mathbf{X}\| + \|\mathbf{W}\| - \mathbf{X}^T \mathbf{W})$ 
  return  $\text{SIGN}(\mathbf{H}\beta)$ 

```

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IX. MATHEMATICAL TYPOGRAPHY AND WHY IT MATTERS

Typographical conventions for mathematical formulas have been developed to **provide uniformity and clarity of presentation across mathematical texts**. This enables the readers of those texts to both understand the author's ideas and to grasp new concepts quickly. While software such as L^AT_EX and MathType[®] can produce aesthetically pleasing math when used properly, it is also very easy to misuse the software, potentially resulting in incorrect math display.

IEEE aims to provide authors with the proper guidance on mathematical typesetting style and assist them in writing the best possible article. As such, IEEE has assembled a set of examples of good and bad mathematical typesetting [1]–[5].

Further examples can be found at <http://journals.ieeeauthorcenter.ieee.org/wp-content/uploads/sites/77/IEEE-Math-Typesetting-Guide-for-Latex-Users.pdf>

A. Display Equations

The simple display equation example shown below uses the “equation” environment. To number the equations, use the \label macro to create an identifier for the equation. LaTeX will automatically number the equation for you.

$$x = \sum_{i=0}^n 2iQ. \quad (34)$$

is coded as follows:

```

\begin{equation}
\label{deqn_ex1}
x = \sum_{i=0}^n 2\{i\} Q.
\end{equation}

```

To reference this equation in the text use the \ref macro. Please see (34)

is coded as follows:

```
Please see (\ref{deqn_ex1})
```

B. Equation Numbering

Consecutive Numbering: Equations within an article are numbered consecutively from the beginning of the article to the end, i.e., (1), (2), (3), (4), (5), etc. Do not use roman numerals or section numbers for equation numbering.

Appendix Equations: The continuation of consecutively numbered equations is best in the Appendix, but numbering as (A1), (A2), etc., is permissible.

Hyphens and Periods: Hyphens and periods should not be used in equation numbers, i.e., use (1a) rather than (1-a) and (2a) rather than (2.a) for subequations. This should be consistent throughout the article.

C. Multi-Line Equations and Alignment

Here we show several examples of multi-line equations and proper alignments.

A single equation that must break over multiple lines due to length with no specific alignment.

The first line of this example

The second line of this example

The third line of this example (35)

is coded as:

```
\begin{multline}
\text{The first line of this example}\\
\text{The second line of this example}\\
\text{The third line of this example}
\end{multline}
```

A single equation with multiple lines aligned at the = signs

$$a = c + d \quad (36)$$

$$b = e + f \quad (37)$$

is coded as:

```
\begin{align}
a &= c+d \\
b &= e+f
\end{align}
```

The align environment can align on multiple points as shown in the following example:

$$x = y \quad X = Y \quad a = bc \quad (38)$$

$$x' = y' \quad X' = Y' \quad a' = bz \quad (39)$$

is coded as:

```
\begin{align}
x &= y & X &= Y & a &= bc \\
x' &= y' & X' &= Y' & a' &= bz
\end{align}
```

D. Subnumbering

The amsmath package provides a subequations environment to facilitate subnumbering. An example:

$$f = g \quad (40a)$$

$$f' = g' \quad (40b)$$

$$\mathcal{L}f = \mathcal{L}g \quad (40c)$$

is coded as:

```
\begin{subequations}\label{eq:2}
\begin{align}
f &= g \label{eq:2A} \\
f' &= g' \label{eq:2B} \\
\mathcal{L}f &= \mathcal{L}g \label{eq:2C}
\end{align}
\end{subequations}
```

E. Matrices

There are several useful matrix environments that can save you some keystrokes. See the example coding below and the output.

A simple matrix:

$$\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \quad (41)$$

is coded as:

```
\begin{equation}
\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix}
\end{equation}
```

A matrix with parenthesis

$$\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad (42)$$

is coded as:

```
\begin{equation}
\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}
\end{equation}
```

A matrix with square brackets

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (43)$$

is coded as:

```
\begin{equation}
\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}
\end{equation}
```

A matrix with curly braces

$$\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix} \quad (44)$$

is coded as:

```
\begin{equation}
\begin{Bmatrix} 1 & 0 \\ 0 & -1 \end{Bmatrix}
\end{equation}
```

A matrix with single verticals

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} \quad (45)$$

is coded as:

```
\begin{equation}
\begin{vmatrix} a & b \\ c & d \end{vmatrix}
\end{equation}
```

A matrix with double verticals

$$\left\| \begin{matrix} i & 0 \\ 0 & -i \end{matrix} \right\| \quad (46)$$

is coded as:

```
\begin{equation}
\begin{Vmatrix} i & 0 \\ 0 & -i \end{Vmatrix}
\end{equation}
```

F. Arrays

The array environment allows you some options for matrix-like equations. You will have to manually key the fences, but there are other options for alignment of the columns and for setting horizontal and vertical rules. The argument to array controls alignment and placement of vertical rules.

A simple array

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \quad (47)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27 \\
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

A slight variation on this to better align the numbers in the last column

$$\left(\begin{array}{cccc} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \quad (48)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{cccc}
a+b+c & uv & x-y & 27 \\
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

An array with vertical and horizontal rules

$$\left(\begin{array}{c|c|c|c} a+b+c & uv & x-y & 27 \\ a+b & u+v & z & 134 \end{array} \right) \quad (49)$$

is coded as:

```
\begin{equation}
\left(
\begin{array}{c|c|c|c}
a+b+c & uv & x-y & 27 \\
a+b & u+v & z & 134
\end{array}
\right)
\end{equation}
```

Note the argument now has the pipe “|” included to indicate the placement of the vertical rules.

G. Cases Structures

Many times cases can be miscoded using the wrong environment, i.e., array. Using the cases environment will save keystrokes (from not having to type the \left\lbrace) and automatically provide the correct column alignment.

$$z_m(t) = \begin{cases} 1, & \text{if } \beta_m(t) \\ 0, & \text{otherwise.} \end{cases}$$

is coded as follows:

```
\begin{equation*}
{z_m(t)} =
\begin{cases}
1, & \text{\text{if}} \backslash \beta_m(t), \\
0, & \text{\text{otherwise.}}
\end{cases}
\end{equation*}
```

Note that the “&” is used to mark the tabular alignment. This is important to get proper column alignment. Do not use \quad or other fixed spaces to try and align the columns. Also, note the use of the \text macro for text elements such as “if” and “otherwise.”

H. Function Formatting in Equations

Often, there is an easy way to properly format most common functions. Use of the \ in front of the function name will in most cases, provide the correct formatting. When this does not work, the following example provides a solution using the \text macro:

$$d_R^{KM} = \arg \min_{d_i^{KM}} \{d_1^{KM}, \dots, d_6^{KM}\}.$$

is coded as follows:

```
\begin{equation*}
d_R^{KM} = \underset{\{\text{\arg min}\} \backslash d_1^{KM},
\ldots, d_6^{KM}}{}
\end{equation*}
```

I. Text Acronyms Inside Equations

This example shows where the acronym “MSE” is coded using \text{} to match how it appears in the text.

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

```
\begin{equation*}
\text{\text{MSE}} = \frac{1}{n} \sum_{i=1}^n
(Y_{i} - \hat{Y}_{i})^2
\end{equation*}
```

X. CONCLUSION

The conclusion goes here.

ACKNOWLEDGMENTS

This should be a simple paragraph before the References to thank those individuals and institutions who have supported your work on this article.

APPENDIX

PROOF OF THE ZONKLAR EQUATIONS

Use `\appendix` if you have a single appendix: Do not use `\section` anymore after `\appendix`, only `\section*`. If you have multiple appendixes use `\appendices` then use `\section` to start each appendix. You must declare a `\section` before using any `\subsection` or using `\label` (`\appendices` by itself starts a section numbered zero.)

REFERENCES SECTION

You can use a bibliography generated by BibTeX as a .bbl file. BibTeX documentation can be easily obtained at: <http://mirror.ctan.org/biblio/bibtex/contrib/doc/> The IEEEtran BibTeX style support page is: <http://www.michaelshell.org/tex/ieeetran/bibtex/>

SIMPLE REFERENCES

You can manually copy in the resultant .bbl file and set second argument of `\begin` to the number of references (used to reserve space for the reference number labels box).

REFERENCES

- [1] *Mathematics Into Type*. American Mathematical Society. [Online]. Available: <https://www.ams.org/arc/styleguide/mit-2.pdf>
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BIOGRAPHY SECTION

If you have an EPS/PDF photo (graphicx package needed), extra braces are needed around the contents of the optional argument to biography to prevent the LaTeX parser from getting confused when it sees the complicated `\includegraphics` command within an optional argument. (You can create your own custom macro containing the `\includegraphics` command to make things simpler here.)

If you include a photo:



Michael Shell Use `\begin{IEEEbiography}` and then for the 1st argument use `\includegraphics` to declare and link the author photo. Use the author name as the 3rd argument followed by the biography text.

If you will not include a photo:

John Doe Use `\begin{IEEEbiographynophoto}` and the author name as the argument followed by the biography text.